Fall 2015 / EC571

Instructor:Ajay Joshi

Final Exam

Date: December 15, 2015

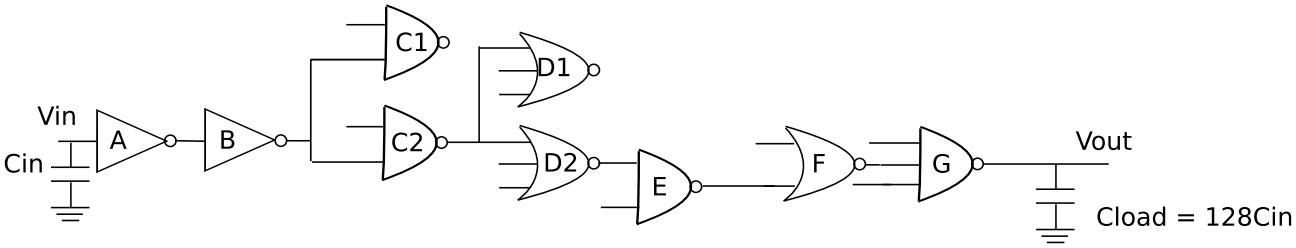
Name : _____

BU ID : _____

Question	Maximum points	Points scored
1	20	
2	20	
3	20	
4	20	
5	20	
6	20	
7	20	
Bonus	5	
Total	100	

1. The duration of the exam is 2 hours.
2. The maximum grade is 100 points.
3. Please write your name on all pages.
4. Attempt any 5 of the 7 questions. Please indicate the question number that should not be graded.
5. The solutions should be written in the spaces provided on the test sheet.
6. If you need to use extra sheets of paper, write your name and BU ID on it, and clearly indicate the question number.
7. Material sharing is not allowed.
8. Calculators are allowed.
9. Boston University Academic Code of Conduct applies.

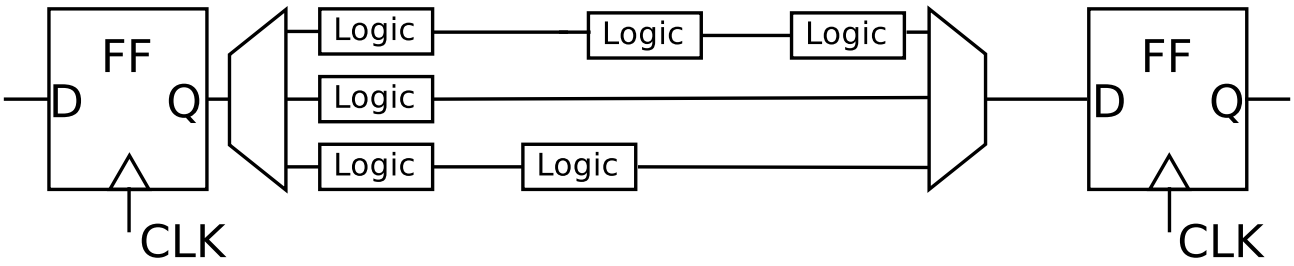
Question 1



For the circuit shown in the above figure, using the logical effort technique, calculate the input width of transistors for each gate in the path such that total delay of the path is minimized. You can assume that gates C1 and C2 are identical and D1 and D2 are identical. Assume Cin = 4 fF, Cg = 1 fF/u. (20 points)

Question 2

Consider the circuit shown below. Assume a unit clk-to-q delay for the register, three units of delay for the multiplexer, two units delay for the demultiplexer and two units of delay for each logic block. The registers are positive-edge triggered and have a setup time of one unit.

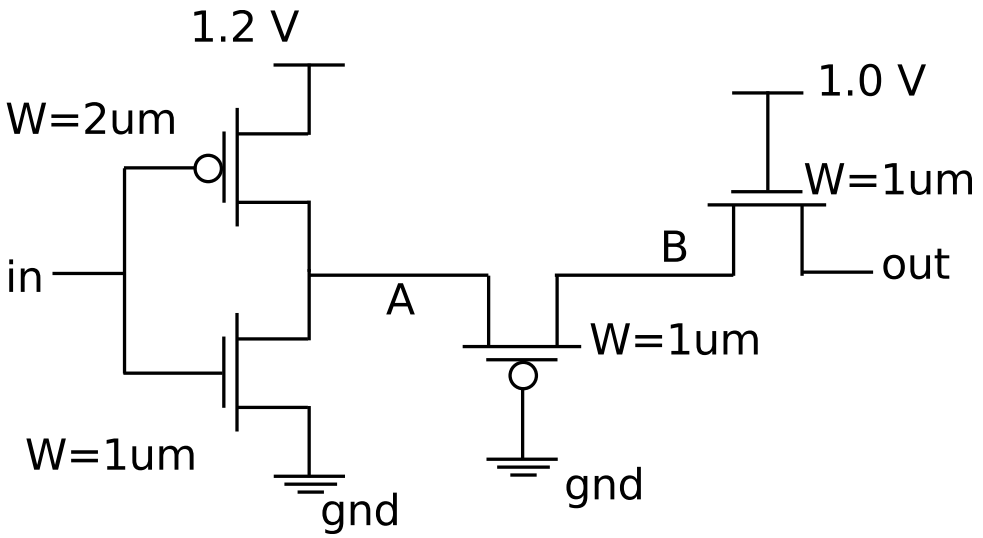


1. Determine the minimum clock period for the above circuit. (7 points)

2. Assuming a hold time of 1 time unit, what is the maximum positive clock skew that can be tolerated by this circuit? (7 points)

3. If the clock input at both FFs has a jitter of ± 2 unit, what would be the minimum clock period of the circuit? Assume zero clock skew (6 points)

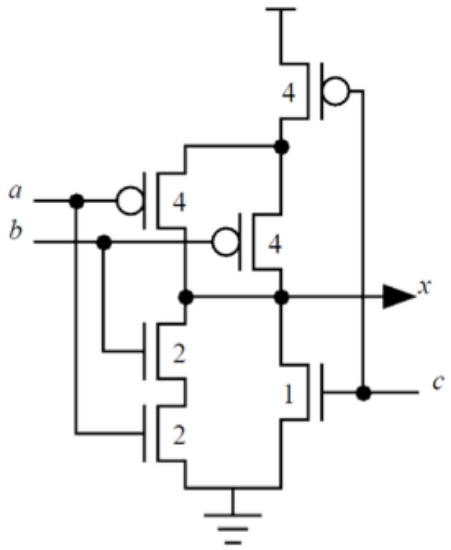
Question 3



Using the simple switch model of the transistors with $R_{sq}(NMOS)$ = 13 Kohm/sq, $R_{sq}(PMOS)$ = 26 Kohm/sq and $V_{th,N}$ = 0.5 V and $V_{th,P}$ = -0.5 V, draw the VTC of the circuit shown above. Explain your answer in detail (20 points).

Question 4

1. For a AOI gate shown below, calculate the worst case high-to-low delay and low-to-high delay using the following parameters - W = 1.5 micron, L = 0.25 micron, $R_{sq,nmos}$=13 Kohm/sq, $R_{sq,pmos}$=26 Kohm/sq, C_g = 1.5 fF/micron, C_d = 2 fF/micron. Assume the output x of AOI drives a load C_L=50 fF. The values next to the transistors indicate relative sizing and W corresponds to the unit size. Consider all internal capacitances. (10 points).



2. Implement $f(A,B,C) = \overline{A \cdot B + C}$ using pseudo-NMOS logic and pass transistor logic (including the inverter at the end). (10 points)

Question 5

Design the static complementary CMOS implementation of an encoder circuit represented by the truth table shown below. Design the circuit for minimum delay (assuming a stage effort of 4) and driving a load of 10 fF. As part of the design you need to determine the width of all transistors. You can use the following transistor parameters for the design - L = 0.25 micron, minimum transistor width = 0.3 micron, $R_{sq}(NMOS)$ = 13 Kohm/sq, $R_{sq}(PMOS)$ = 26 Kohm/sq, C_g = 1 fF/micron, C_d = 1.5 fF/micron. You can assume that you have both complemented and non-complemented inputs and that input signals have very small rise and fall times. *Hint: Determine the Boolean expression for each output column of the encoder circuit and use the Boolean expression to design the circuit. Use logical effort technique to determine the transistor sizes.* (20 points)

Table 1: Truth table for Encoding

Conventional Binary Encoded Value	New Encoded Value
$b_3\ b_2\ b_1\ b_0$	$e_1\ e_0$
0 0 0 0	1 1
0 0 0 1	1 0
0 0 1 0	1 X
0 0 1 1	1 0
0 1 0 0	0 0
0 1 0 1	0 0
0 1 1 0	0 X
0 1 1 1	0 0
1 0 0 0	0 X
1 0 0 1	X 1
1 0 1 0	0 X
1 0 1 1	X 1
1 1 0 0	0 0
1 1 0 1	1 X
1 1 1 0	0 0
1 1 1 1	1 1

Question 6

For the circuit shown below, the signal waveforms per cycle at nodes A, B, C, D and E are shown next to it. You can assume the these signal waveforms repeat every cycle. You can assume the following device parameters: V_T = 0.3 V, R_{ON} = 20 kΩ and R_{OFF} = infinity, V_{dd} = 1.0 V.

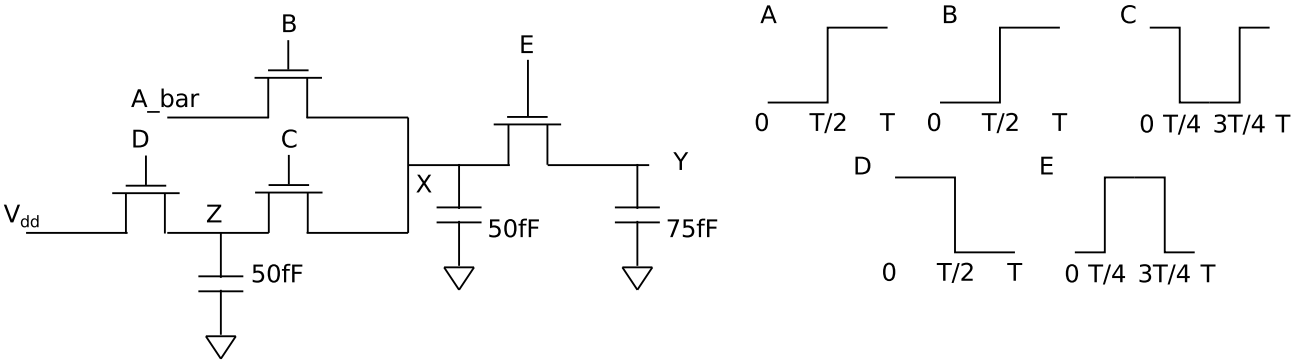


Figure 1

1. Draw the timing waveforms of the signals on nodes X, Y and Z for a time of 0 to 4τ. You can assume the signals at nodes A, B, C, D and E are switching between 0 and 1.0 V, and each quarter of the period is long enough to charge/discharge the capacitors. (8 points)

2. Calculate the total dynamic energy consumed during each of the 4 cycles by the capacitors at nodes X, Y and Z. (8 points)

3. Draw the timing waveforms of the signals on nodes X, Y and Z for a time of 0 to 4τ if all the NMOS transistors were replaced with transmission gates. You can assume the signals at nodes A, B, C, D and E are switching as shown in the above figure, and between 0 and 1.0 V. (4 points)

Question 7

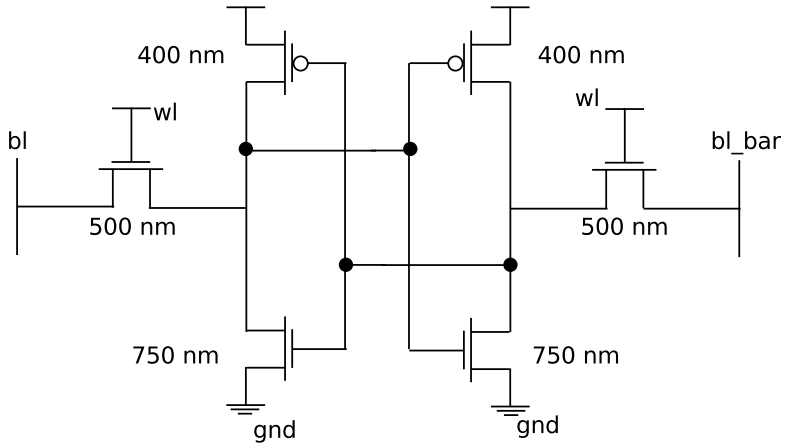


Figure 2

For this problem we are considering a 256 x 256 cell SRAM array (each wordline drives 256 cells and each bitline\bitline_bar has 256 cells on it).The dimensions of each cell are 1 μm tall x 1.5 μm wide. Assume both wordline and bitline\bitline_bar are 500 nm wide. Assume $C_g = C_d$ = 1.5 fF/μm, $R_{sq,NMOS}$ = 13 kΩ/sq, $R_{sq,PMOS}$ = 26 kΩ/sq, R_w = 2 Ω/μm and C_w = 0.2 fF/μm

1. What is the total capacitance and resistance of the bitline\bitline_bar and wordline?(6 points)

2. What is the total dynamic energy consumed in the bitline, bitline_bar and wordline during a read operation? Assume both bitline and bitline_bar are discharged before the read operation and V_{dd} = 1.2 V. (7 points)

3. Assume the the wordline is driven by an inverter. What should be the width of the NMOS and PMOS of the inverter to have a fanout of 4 delay in charging up the wordline? (7 points)

Bonus question (5 points)
Prof. Joshi wants to escape the Boston winter during the break between Fall and Spring semester. He is considering traveling to the southern hemisphere. Name atleast 5 places in the southern hemisphere that he could visit during the break.